

Note: - Attempt all questions in the given order only. Missing data/information (if any), may be suitably assumed & mentioned in the answer. All questions carry equal marks.

Q. No.	Question	Marks	CO																
1a ✓	Describe random variable, the cumulative distribution function (CDF) and the probability density function (pdf). Discuss the relation between the CDF and the pdf.	2 / 2.5	CO1																
1b ✓	Let Y be a continuous random variable with the probability density function (pdf) given as $f(y) = \text{Exp}(- y)$. Find out (i) $P(-1 \leq Y \leq 1)$ (ii) $\text{Variance}(Y)$	2 / 2.5	CO2																
2a	Consider two discrete random variables X and Y with the following joint probability distribution: <table border="1"> <thead> <tr> <th>$P(X, Y)$</th><th>$Y=0$</th><th>$Y=1$</th><th>$Y=2$</th></tr> </thead> <tbody> <tr> <td>$X=0$</td><td>0.1</td><td>0.2</td><td>0.1</td></tr> <tr> <td>$X=1$</td><td>0.1</td><td>0.1</td><td>0.1</td></tr> <tr> <td>$X=2$</td><td>0.1</td><td>0.1</td><td>0.1</td></tr> </tbody> </table> (i) Find marginal distributions of X and Y (ii) Find $E(XY)$	$P(X, Y)$	$Y=0$	$Y=1$	$Y=2$	$X=0$	0.1	0.2	0.1	$X=1$	0.1	0.1	0.1	$X=2$	0.1	0.1	0.1	2 / 2.5	CO2
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2b ✓	The pdf of a random variable X is given by $f_X(x) = 1$ for $0 \leq x \leq 1$, and zero elsewhere. Calculate its variance.	2 / 2.5	CO2																
3a ✓	Find characteristic function of Poisson distribution	2 / 2.5	CO3																
3b ✓	Describe Gaussian distribution. Derive its mean.	2 / 2.5	CO2																
4a ✓	Let $Z=X+Y$, where X and Y are independent random variables, find out the PDF of Z in terms of PDFs of X and Y .	2 / 2.5	CO2																
4b ✓	Explain conditional distribution and conditional expectation.	2 / 2.5	CO2																
5a ✓	Let $Y = aX+b$, X and Y are random variables, a and b are the constants. Derive the PDF of Y in terms of the PDF of X .	2 / 2.5	CO2																
5b ✓	Two random variables- X and Y are independent. Prove that they are uncorrelated.	2 / 2.5	CO3																